B.TECH 4-YEAR CURRICULUM & SYLLABUS FRAMEWORK (ALL 7 DISCIPLINES)

Course Structures, Core Disciplines, Professional Electives & Laboratory Frameworks

Ref: RGUKT/ACAD/CURR-2024 | **Effective: 2024-2028 Batch** | **Authority: Board of Studies, RGUKT-AP**

1. Engineering Disciplines Offered Across Campuses

The 4-Year B.Tech engineering curriculum is structured across seven specialized departments according to AICTE model guidelines and RGUKT statutory ordinances:

- **Computer Science & Engineering (CSE):** Algorithms, Data Structures, AI/ML, Cloud Computing, Distributed Systems, DBMS, Cyber Security, Web Tech.
- **Electronics & Communications Engineering (ECE):** VLSI Design, Embedded Systems, Microprocessors, Wireless & 5G Communication, DSP, RF Microwave.
- **Electrical & Electronics Engineering (EEE):** Power Systems, Power Electronics, Smart Grids, Electric Vehicle Drives, Renewable Energy, Control Systems.
- **Mechanical Engineering (MECH):** CAD/CAM, Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Thermodynamics, Robotics, Metrology.
- **Civil Engineering (CIVIL):** Structural Analysis, Transportation & Highway Design, Geotechnical Engineering, Water Resources, GIS & Surveying.
- **Chemical Engineering (CHEM):** Chemical Reaction Engineering, Heat & Mass Transfer Operations, Process Dynamics & Control, Green Chemistry.
- **Metallurgical & Materials Engineering (MME):** Physical Metallurgy, Nanomaterials, Advanced Characterization (SEM/XRD), Phase Transformations, Corrosion.

2. Minor Degrees & Honors Provision

Students with CGPA >= 8.0 and no active backlogs may earn an additional Minor Degree (20 credits) in areas such as Business Management, Artificial Intelligence, Computational Mathematics, or an allied engineering stream.